

Weekly Progress Report

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Domain: Python Development Internship

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Week Ending: 01

I. Overview:-

This week my primary focus was on building four core Python projects from scratch: a URL Shortener, a File Organizer, a Password Manager, and a Quiz Game. I set up my Python environment (Python 3.14, VS Code) and installed the required libraries. I spent time understanding how web frameworks, databases, and encryption work by directly applying them to these projects.

II. Achievements:-

1. Environment & Technology Setup:-

- Installed Python 3.14 and VS Code Set up a dedicated project folder (D:\Internship project) to keep all four projects organized.
- Installed external libraries like Flask (for the web app) and cryptography (for encryption) using pip.
- Learned how to use SQLite3 for local data storage across three of the projects.

2. Python Project Contributions:-

Name of the project:- URL Shortener

- Built a fully functional web application using Flask. It takes a long URL generates a unique short code, and redirects users to the original URL.
- I tested it by shortening 10 different URLs and all redirected correctly. I also tested the Command-Line Interface (CLI) mode by using the shorten and expand commands in the terminal.

Name of the project:- File Organizer

- Created a command-line utility that scans a directory categorizes files (Images, Documents, Videos, etc.) based on extensions, and moves them into separate folders.
- I organized my entire Downloads folder, which contained over 100 files, without any data loss.

Name of the project:- Password Manager

- Developed a secure password storage system with Master Password authentication and Fernet encryption to keep passwords safe.
- I stored 5 different account passwords and successfully retrieved them using the master password.

Name of the project:- Quiz Game

- Designed an interactive quiz application that reads questions from a JSON file. It supports MCQ and True/False and saves high scores.
- I added 15 custom questions about Python programming and played it with my friends.

3. Learning Python:-

- Gained hands-on experience with essential Python libraries: Flask, SQLite3, Cryptography (Fernet), OS/Shutil and JSON.
- Applied theoretical Python knowledge to solve real-world problems like file handling, data persistence, and secure storage.

III. Challenges:-

- I faced a Module Not Found Error : No module named flask when I first ran the URL Shortener.
- Resolution: I simply ran pip install flask in the terminal, and it started working. I also learned that ignoring the PATH warning is fine when running scripts via python filename.py.

IV. Learning Resources:-**1. Official Documentation & References:**

- Used the Flask Official Documentation for routing and handling web requests.
- Referred to the Python Standard Library (os, shutil, sqlite3, json) extensively.
- Used the Cryptography.io documentation to understand Fernet encryption.

2. Python Learning Resources:-

- Watched YouTube tutorials on Flask and File Handling to understand the concepts visually.
- Used Stack Overflow and ChatGPT to debug specific errors (like SQL Operational Error and Fernet key issues).
- Practiced by modifying the sample codes to fit my project needs.

V. Next Week's Goals:-

1. Project Enhancement & Deployment:

- I am planning to deploy the URL shortener on Render.com so that my friends can use it online.
- I will add a timer feature to the quiz game to make it more challenging.

2. Documentation & Code Quality:

- Start preparing the final project documentation (User Manual) in a Word/PDF file, including installation steps and screenshots.
- Refactor my code to follow PEP 8 guidelines and add proper comments.

3. Continuous Learning:

- Explore how to build REST APIs using Flask so the URL Shortener can be used by other applications.

VI. Additional Comments:

- I spent about 5 hours every day. My favorite moment was when I successfully ran the URL shortener on localhost and viewed the web page.
- I have uploaded all the code to my personal GitHub repository for backup.